

**BEFORE
PENNSYLVANIA PUBLIC UTILITY COMMISSION**

In Re: Application of Aqua Pennsylvania, Inc. for Approval of the alteration of the crossing (U.S DOT – 589 566 C) where Lee St crosses at grade tracks operated by SEPTA at milepost 0012.35 in Whitemarsh Township in Montgomery County, Pennsylvania.

Application Docket No.:

To the Pennsylvania Public Utility Commission:

1. The name and address of applicant is Aqua Pennsylvania, Inc., 762 West Lancaster Avenue, Bryn Mawr, Pennsylvania 19010.
2. The name and address of the applicant's attorney is Frances P. Orth, Esquire, 762 West Lancaster Avenue, Bryn Mawr, Pennsylvania 19010.
3. Aqua Pennsylvania, Inc. is the local public water supplier.
4. By this application, approval is sought to install a 12-inch PVC pipe in a 20-inch steel casing in the road in SEPTA ROW at the crossing of Lee St located at milepost 0012.35. The crossing number is 589 566 C operated by SEPTA in Whitemarsh Township in Montgomery County, Pennsylvania.
5. To the best of the applicant's knowledge, the name and addresses of the persons, parties and entities potentially concerned in or affected by the proposed construction are:

PECO 2301 Market Street P.O. Box 8699 Philadelphia, PA 19101 Attn: Engineering Department	AT&T 1100 3rd Ave Altoona, PA 16602 Attn: Patt Sutton
SEPTA Facilities Engineering Department 1234 Market Street Philadelphia, PA 19107 Attn: Cliff Kim	PAPUC Railroad Crossing Applications Penn DOT Right of Way and Utilities Division Penn Dot Bureau of Design P.O. Box 3362 Attn: Mark Chappell PE Chief
PennDOT District 6-0 7000 Geerdes Boulevard King of Prussia, PA 19406-1525 Attn: Mary Ann Lang	Karen Cummings Assistant Counsel in Charge Department of Transportation Office of the Chief Counsel 9 th Floor P.O. Box 8212 Harrisburg, PA 17105-8212
Crown Castle 1500 Corporate Drive Canonsburg, PA 15317 Attn: John Smith	Comcast 1250 Haddonfield-Berlin Road Cherry Hill, NJ 08034 Attn: John Cedrick Salvatierra

Verizon Business Formerly MCI 7000 Weston Pkwy Cary, NC 27513 Attn: Victor Wood	Verizon Business Formerly MCI 400 International Pkwy Richardson, TX. 75081 Attn: Robert Butler
Lumen Formerly Level 3 1025 Eldorado Blvd Broomfield, CO 80021 Attn: Lumen Operator Personnel	Montgomery County One Montgomery Plaza Suite 800 Norristown, PA 19404 Attn: Lee Soltysiak
Whitemarsh Township Public Works 616 Germantown Pike Lafayette Hills, PA. 19444 Attn: John Fields	Whitemarsh Township Authority 2015 Joshua Rd Bldg 100 Lafayette Hill, PA 19444 Attn: Tom Bonjo
Total Engineering & Consulting Services LLC 1547 DeKalb St Norristown PA 19401 Attn: Elizabeth Sobel	Aqua Pennsylvania 762 Lancaster Ave Bryn Mawr, PA 19010 Attn: Thomas Waddy

6. The cost of this portion of the project is estimated to be \$140,000.00. The project will be entirely funded and maintained by Aqua Pennsylvania, Inc.
7. The alteration is necessary or proper because it will improve water distribution operations and maintenance.

Wherefore, applicant requests the Commission approve this highway railroad crossing application.

RESPECTFULLY SUBMITTED:

Aqua Pennsylvania, Inc.

Date: 5/22/2025

By: 
Scott R. Steffy, Engineer III

VERIFICATION

I, Elizabeth Sobel, Vice President at Total Engineering and Consulting Services, LLC., hereby state that the facts above set forth are true and correct (or are true and correct to the best of my knowledge, information and belief) and that I expect to be able to prove the same at a hearing held in this matter. I understand that the statements herein are made subject to the penalties of 18 Pa. C.S. § 4904 (relating to unsworn falsification to authorities).

Date: 5/22/2025

A handwritten signature in black ink, appearing to read "Elizabeth Sobel", written over a horizontal line.

Elizabeth Sobel

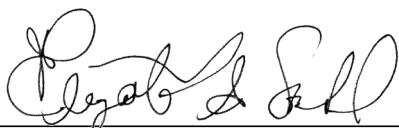
CERTIFICATE OF SERVICE

I, Elizabeth Sobel, hereby certify that I have this day served a true copy upon the participants, listed below, in accordance with the requirements of 52 PA Code Section 1.54 regarding the Application of Aqua Pennsylvania, Inc. for a water main installation at the crossing (U.S. DOT – 589 566 C) where Lee St crosses above grade tracks operated by SEPTA located at milepost 0012.35 at crossing 589 566 C in Whitemarsh Township in Montgomery County, Pennsylvania.

By U.S. 1st Class Mail

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Dated: 05/22/2025



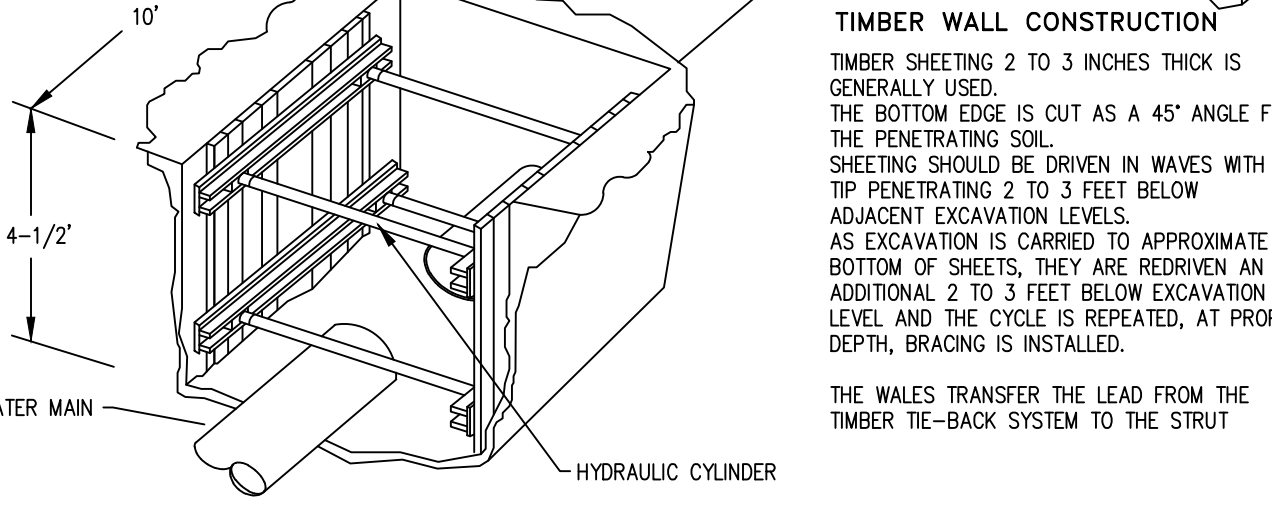
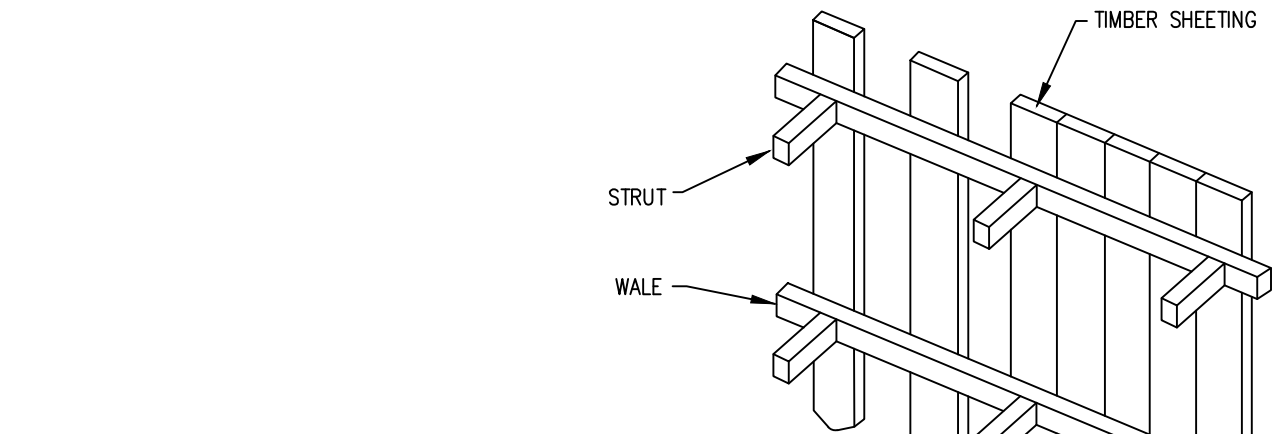
Elizabeth Sobel

SEPTA CONSTRUCTION REQUIREMENTS

1. OSHA RULE AND REGULATIONS MUST BE ADHERED TO FOR SAFETY IN EXCAVATIONS, THE DESIGN AND THE SHORING OF THE BORING PITS. ADDITIONALLY, PITS SHALL BE FENCED WITH ADEQUATE LIGHTING FOR PUBLIC SAFETY.
2. THE TRACK AND RIGHT-OF-WAY MUST BE PROTECTED FROM ANY EROSION CAUSED BY THE PROJECT FIELD WORK.
3. SUPPORT OF EXCAVATION (SOE) DESIGN IS THE RESPONSIBILITY OF THE CONTRACTOR. SOE DESIGN TO BE SUBMITTED TO SEPTA FOR REVIEW PRIOR TO CONSTRUCTION.
4. PROVIDE EVIDENCE THAT SEPTA'S INSURANCE REQUIREMENTS, AS DETERMINED BY SEPTA'S RISK MANAGEMENT OFFICE, HAVE BEEN MET.
5. ALL PERSONNEL INVOLVED IN THE FIELD WORK PORTION OF THE PROJECT, AND WHO'S DUTIES MIGHT INVOLVE ENTERING THE SEPTA RIGHT-OF-WAY, WILL BE REQUIRED TO ATTEND A RWP TRAINING IN ORDER TO OBTAIN SAFETY CERTIFICATION, AS MANDATED BY SEPTA AND THE FEDERAL RAILROAD ADMINISTRATION. AQUA WILL COORDINATE WITH SEPTA PM TO SET UP SEPTA RWP TRAINING (THROUGH RAILPROS) FOR ALL WORKERS WHO WILL BE ON SEPTA PROPERTY.
6. PROVIDE NOTIFICATION PRIOR TO THE START OF CONSTRUCTION, INCLUDING UPDATED PROJECT SCHEDULES.
7. STAINLESS STEEL CASING SPACERS SPECIFIED SHALL HAVE TWO PIECE STAINLESS BAND WITH POLYETHYLENE RUNNER HEIGHT OF ONE AND A HALF INCHES. RECOMMENDED PLACEMENT OF SPACERS ARE 3 PER 18-20 FT. LENGTH OF PIPE. SPACERS SHALL BE ADVANCED PRODUCT SYSTEMS, (337) 322-6116 OR APPROVED EQUAL.
8. A RIGHT OF ENTRY PERMIT MUST BE SECURED PRIOR TO START OF CONSTRUCTION.
9. A SIGN IDENTIFYING THE BURIED UTILITY AND EMERGENCY SHUTOFF VALVES WILL BE INSTALLED AT THE PIPE LOCATION. THE SIGN SHALL INCLUDE THE FACILITY OWNER NAME (AQUA) AND CONTACT NUMBER.
10. AQUA WILL PROVIDE A REGISTERED SURVEYOR WHO SHALL ESTABLISH THE TOP-OF-RAIL ELEVATIONS, FOR EACH RAIL, PRIOR TO THE START OF WORK. SUBSEQUENT ELEVATION MEASUREMENTS SHALL BE TAKEN DURING AND AT THE COMPLETION OF WORK, IN ORDER TO MONITOR TRACK CONDITIONS ABOVE THE BORE.
11. RAIL MONITORING SHALL BE REQUIRED. RAIL MONITORING DESIGN AND PLAN SHALL BE COMPLETED BY THE CONTRACTOR OR REGISTERED SURVEYOR AND SUBMITTED TO SEPTA FOR APPROVAL PRIOR TO CONSTRUCTION. AT A MINIMUM, RAIL MONITORING REQUIREMENTS ARE AS FOLLOWS:
 - 11.1. ESTABLISH REMOTE MONITORING POINTS (RMPS) FOR COLLECTION OF DATA.
 - 11.2. PRECONSTRUCTION MONITORING ACTIVITIES TO BEGIN AT LEAST 10 DAYS PRIOR TO START OF CONSTRUCTION
 - 11.3. MONITOR USING TOTAL STATION INSTRUMENT HAVING A MINIMUM ANGULAR ACCURACY OF 1-SECOND AND AN ELECTRONIC DISTANCE MEASUREMENT ACCURACY OF 1.0MM + 2PPM. TOTAL STATION TO LOCATE RMPS ON TRACK TO BE MONITORED
 - 11.4. CONTRACTOR SHALL SPACE THE PRISMS AT 15'-6" INTERVALS
 - 11.5. RMPS SHALL BE EITHER COMMERCIALY AVAILABLE CALIBRATED REFLECTIVE TARGETS OR SMALL PRISMS. ALL TARGETS SHALL BE MOUNTED AT A UNIFORM ELEVATION BELOW THE TOP OF RAIL
 - 11.6. REFLECTIVE TARGETS SHALL BE LESS THAN 3-INCHES SQUARE AND MOUNTED TO THE WEB OF THE RAIL BY ADHESIVE. ALLOWABLE MINIMUM ANGLE FROM INSTRUMENT TO TARGET FACE IS 30 DEGREES.
 - 11.7. PRISMS MOUNTED ON RAIL BRACKETS SHALL REMAIN AT LEAST 1-INCH BELOW TOP OF RAIL HEAD AND MUST NOT INTERFERE WITH TRACK COMPONENTS.
 - 11.8. CONTRACTOR SHALL INCLUDE PRISMS ON THE TRACK(S) THE WORK PASSES BENEATH.
 - 11.9. IF THE PRISMS ENCOUNTER SPECIAL TRACK WORK LOCATIONS (CROSSINGS, TURNOUTS, MITER RAILS), CONTRACTOR SHALL NOTIFY SEPTA TO DETERMINE AN ALTERNATE ARRANGEMENT.
 - 11.10. THE TOTAL COUNT ALONG THE TRACK SHALL EXTEND THROUGH THE ZONE OF INFLUENCE AND EXTEND AN ADDITIONAL FIVE (5) STATIONS, IN EITHER DIRECTION, OUTSIDE OF THE ZONE OF INFLUENCE.
 - 11.10.1. ZONE OF INFLUENCE USING THE DIAMETER OR WIDTH OF THE UNDERGROUND WORK, EXTENDING TO THE SURFACE OF THE GROUND AT THE SOIL ANGLE OF REPOSE. SOIL ANGLE OF REPOSE SHOULD BE TAKEN FROM THE DEPTH OF THE UNDERGROUND WORK TO GROUND LEVEL.
12. TO MINIMIZE THE POTENTIAL OF THE BORE HOLE COLLAPSING, ALL DRILLING AND BACK REAMING OPERATIONS MUST UTILIZE TRAIL RODS.
13. CASING PIPE BORING AND DRILLING OPERATIONS SHALL COMMENCE CONTINUOUSLY WITHOUT STOPPAGE.
14. ALL SPOILS GENERATED FROM THE BORING AND DRILLING OPERATIONS SHALL BE TESTED OR TAKEN TO DEP-APPROVED SITE FOR PROPER DISPOSAL. NO ON-SITE STORAGE OF SPOIL MATERIALS WILL BE PERMITTED WITHIN SEPTA R/W.
15. ALL CONTRACTORS SHALL NOTIFY THE PENNSYLVANIA ONE CALL SYSTEM INCORPORATED AT 1-800-242-1776 FOR BURIED UTILITIES LOCATIONS.

TOOL BOX MEETINGS:

THE CONTRACTOR'S SUPERINTENDENT OR SAFETY OFFICER SHALL HOLD DAILY "TOOL BOX" JOB BRIEFINGS WITH ALL CONTRACTOR PERSONNEL. SUBJECTS, TIME, AND LOCATION CAN BE SET AT THE CONTRACTOR'S CONVENIENCE. SEPTA REQUIRES 3 BUSINESS DAYS PRIOR NOTICE, OR A SCHEDULE OF SUCH TOOL BOX MEETINGS. THIS NOTICE SHALL INCLUDE LOCATION AND AGENDA.



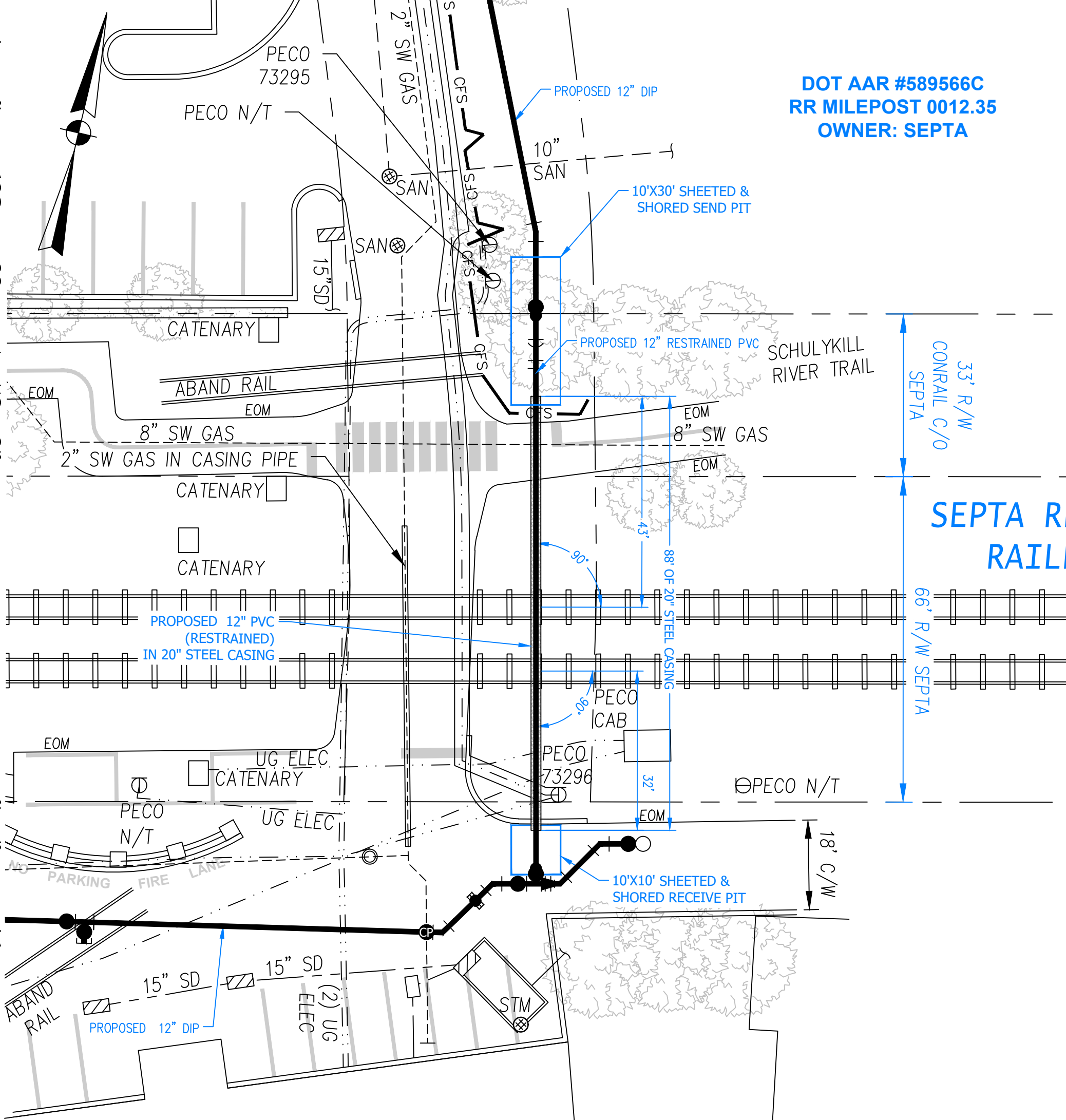
HYDRAULIC SHORING SHOULD BE CHECKED AT LEAST ONCE PER SHIFT FOR LEAKING HOSES AND/OR CYLINDERS, BROKEN CONNECTIONS, CRACKED NIPPLES, BENT BASES, AND ANY OTHER DAMAGED OR DEFECTIVE PARTS

All shoring should be installed from the top down and removed from the bottom up.

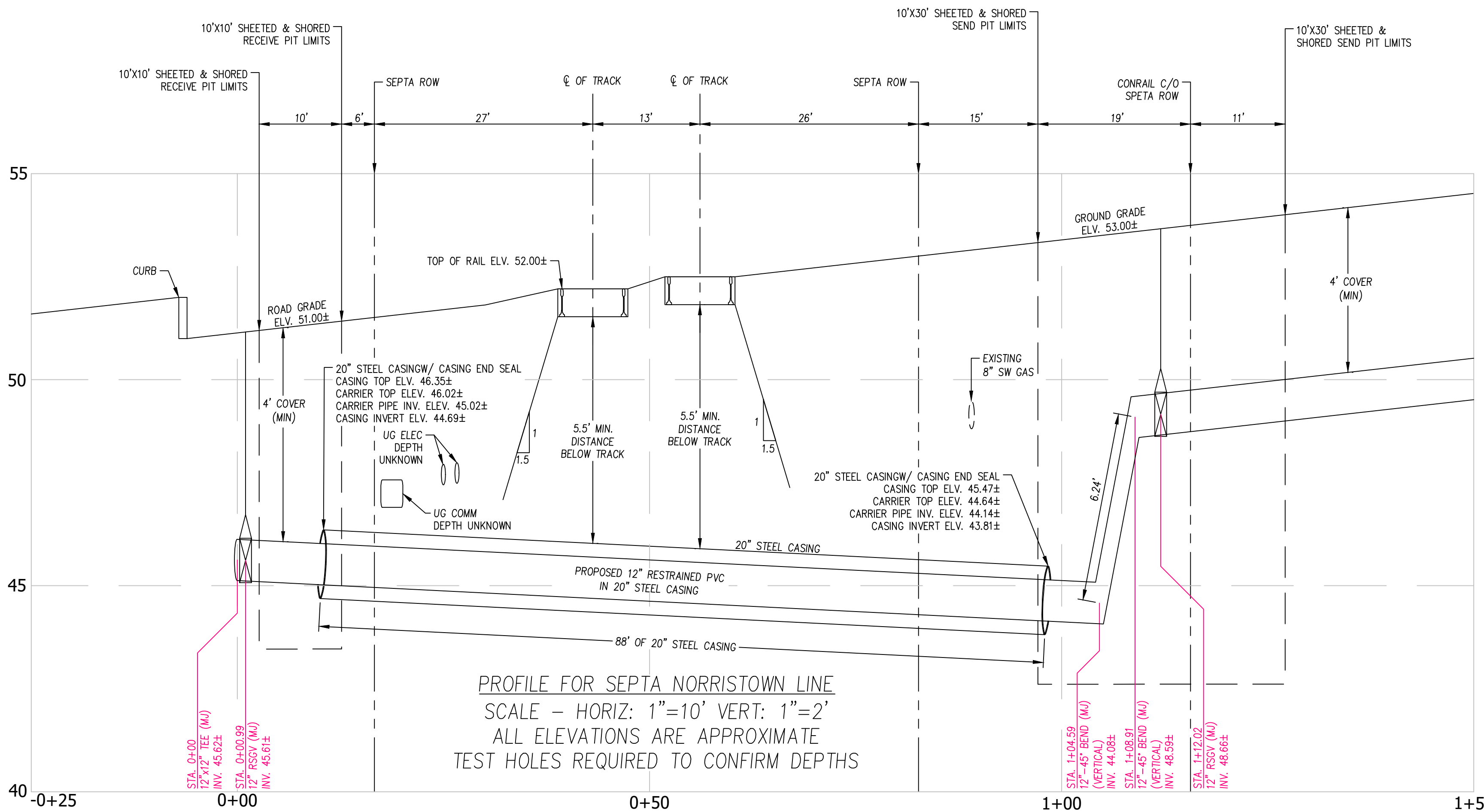
TYPICAL SHEETING AND SHORING

BORING PROCEDURE

1. WORK AREAS WILL BE CLEARED, PRECAUTIONS TO PROTECT EXISTING DRAINAGE AND EROSION CONDITIONS WILL BE TAKEN. APPROPRIATE SIGNS TO DIVERT TRAFFIC WILL BE PLACED IN ACCORDANCE WITH SEPTA AND WHITEMARSH TWP SPECIFICATIONS; NO EQUIPMENT SHALL BE OPERATED ON OR NEAR THE TRACK IN ANY WAY THAT COULD ERODE THE TRACK SHOULDER OR TRACK STRUCTURE. EQUIPMENT IS THEN STAGED ON SITE AND NECESSARY TRAFFIC CONTROLS AND DESIGNATED WORK AREAS PUT IN PLACE.
2. CONSTRUCT ONE (1) 8'X30' LAUNCH PITS AND ONE (1) 10'X10' EXIT PIT. THE LAUNCH PITS WILL BE OUTSIDE OF THE SEPTA RIGHT-OF-WAY. ALL PITS WILL HAVE SHEETED AND SHORED WALLS PER DETAIL (SEE THIS SHEET).
3. INSTALL BORING MACHINE RAILS AND LEVEL WITH FLOOR OF LAUNCH PIT. PLACE THE BORING MACHINE ON THE RAILS.
4. PLACE CASING PIPE ON RAILS AND ATTACH BORING HEAD AND AUGERS TO THE BORING MACHINE.
5. THE TRACKS OF THE RAILROAD WILL BE SURVEYED BEFORE BORING BEGINS.
6. AS THE HOLE IS BORED, SOIL IS REMOVED FROM LAUNCH PIT AND A COMPLETE SECTION OF CASING PIPE IS INSERTED. THE NEXT SECTION OF CASING IS WELDED TO THE EXISTING CASING AND AUGERS ARE ADDED TO THE BORING MACHINE. THIS PROCEDURE IS REPEATED UNTIL THE CASING EXTENDS ACROSS THE RAILROAD RIGHT-OF-WAY INTO THE EXIT PIT.
7. DURING THE BORING PROCEDURE THE TRACKS WILL BE SURVEYED TO CHECK FOR ANY CHANGE IN ELEVATION.
8. AFTER THE CASING IS INSERTED THE CASING SPACERS ARE ATTACHED TO THE CARRIER PIPE PER DETAIL (SEE THIS SHEET).
9. THE CARRIER PIPE AND SPACERS ARE INSERTED IN THE CASING PIPE UNTIL IT REACHES THE OTHER SIDE OF THE RAILROAD RIGHT-OF-WAY AND/OR THE EXIT PIT.
10. THE CASING PIPE IS SEALED PER DETAIL, (SEE THIS SHEET), THE EXIT PIT AND TRENCH ARE BACKFILLED AND RESTORED TO MATCH EXISTING CONDITIONS.
11. THE BORING MACHINE AND RAILS ARE REMOVED AND THE LAUNCH/EXIT PITS ARE BACKFILLED AND RESTORED TO EXISTING CONDITIONS. BACKFILL AND RESTORATION TO BE IN ACCORDANCE WITH WHITEMARSH TWP SPECIFICATIONS AND REQUIREMENTS. SIGNS IDENTIFYING THE BURIED UTILITY SHALL BE PLACED AT THE PIPE LOCATION OUTSIDE RAILROAD RIGHT-OF-WAY.
12. THE TRACK OF THE RAILROAD IS THEN SURVEYED FOR ANY ELEVATION CHANGES. WHEN DISTURBED AREAS ARE RESTORED TO ORIGINAL CONDITION, TRAFFIC CONTROL AND EQUIPMENT IS REMOVED.

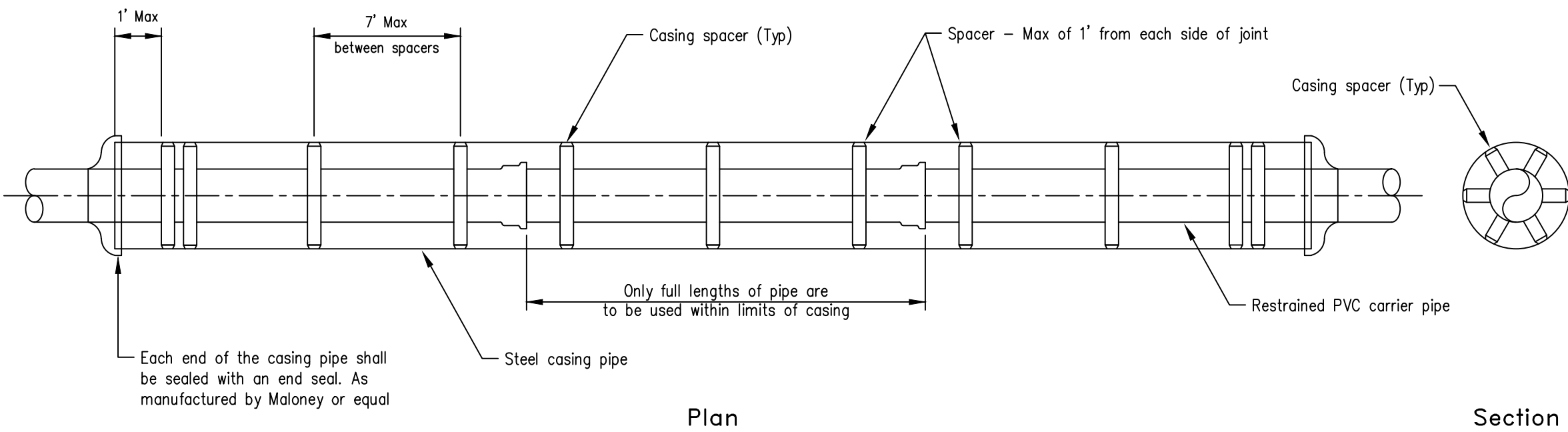


PLAN FOR SEPTA MANAYUNK/NORRISTOWN LINE
SCALE - 1"=20'



PROFILE FOR SEPTA NORRISTOWN LINE
SCALE - HORIZ: 1"=10' VERT: 1"=2'
ALL ELEVATIONS ARE APPROXIMATE
TEST HOLES REQUIRED TO CONFIRM DEPTHS

TYPICAL PVC WATER MAIN WITH STEEL CASING AND SPACERS



Pipe Crossing Data	Carrier Pipe	Casing Pipe
Contents to be handled	Water	None
Operating Pressure (psi)	110	N/A
Nominal pipe size (")	12"	20"
O.S. Diameter (")	13.20"	20.00"
I.S. Diameter (")	11.20"	19.00"
Wall Thickness (")	0.943" (Min.)	0.50"
Weight per Foot (lbs)	25.1	104.3
Material	Polyvinyl Chloride (PVC)	Steel
Process of Manufacture	Extrusion - Molded	Welded Round Pipe
Specification	AWWA C900	ASTM A53 & AWWA C200
Grade or Class	SDR 14	A53 GRADE B
Working Pressure (psi)	305	N/A
Type of Joint	Restrained	Welded
Type of Coating	N/A	Bitumastic
Details of Cathodic Protection	N/A	Wrap or Coating
Details of Seal or Protection at Ends of Coating	N/A	Flexible oil/water resistant rubber end seals
Method of Installation	Pushing	Jacking Boring
Specific Min. Yield Strength	15,000	35,000 PSI

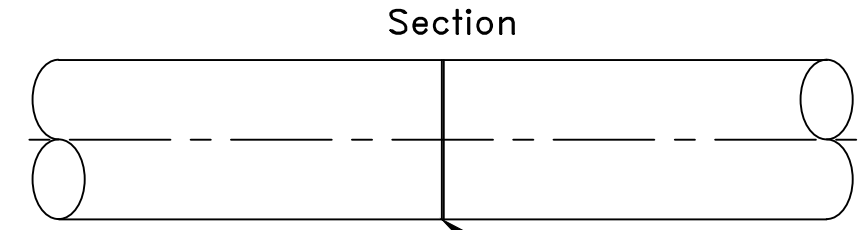
STEEL CASING PIPE NOTES:

1. ALL CASING PIPE UP TO 24" DIAMETER SHALL BE ASTM A53 GRADE B, 0.50" WALL THICKNESS, STANDARD WEIGHT, MANUFACTURED IN ACCORDANCE WITH AWWA C200.
2. CASING PIPE SHALL BE REQUIRED FOR WATER MAINS CROSSING OVER AND UNDER HIGHWAYS, RAILROADS, OR ANY OTHER LOCATION WHERE A LEAKING PIPE CAN CREATE A HAZARDOUS CONDITION; IN GENERAL, CASING PIPE IS NOT REQUIRED FOR MAINS CROSSING BODIES OF WATER.
3. ALL EXPOSED CASING PIPE SHALL BE SHOP BLASTED AND COATED EXTERNALLY IN ACCORDANCE WITH THE STATE DOT PAINT SPECIFICATION FOR STRUCTURAL STEEL.
4. ALL STEEL PIPE BELOW GRADE SHALL BE WRAPPED WITH PROTECTO-WRAP 310, COLD APPLIED PIPE TAPE OR CARBOLINE COATING. IN CASES WHERE STEEL PIPE RUNS ABOVE GRADE, COATING SHALL EXTEND TO A MINIMUM OF 18" BEYOND FINISHED GRADE. IF APPLICABLE, REFER TO THE CATHODIC PROTECTION PLAN FOR ADDITIONAL CORROSION PROTECTION MEASURES.
5. IN CASES WHERE STEEL PIPE IS TO BE WELDED IN THE FIELD, PIPE SHALL BE ORDERED WITHOUT COATING 6" FROM EACH END TO ALLOW FOR BEVELING AND WELDING IN THE FIELD.
6. ALL WELDERS SHALL BE CERTIFIED IN ACCORDANCE WITH THE API 1104 CODE OR AN APPROVED EQUAL. CONTRACTOR SHALL PROVIDE OWNER WITH EVIDENCE OF CURRENT WELDER QUALIFICATION PRIOR TO THE COMMENCEMENT OF CONSTRUCTION.
7. ALL PRODUCTION WELDS SHALL BE PERFORMED, TESTED AND INSPECTED IN CONFORMANCE WITH THE CURRENT EDITION OF AWWA C206.
8. ANY WELDER WHO MAKES A PRODUCTION WELD THAT FAILS TO COMPLY WITH THE REQUIREMENTS OF AWWA C206 MAY BE IMMEDIATELY DISQUALIFIED FROM CURRENT AND/OR FUTURE WORK AT THE DISCRETION OF THE OWNER.
9. STEEL CASING SHALL BE SO CONSTRUCTED AS TO PREVENT LEAKAGE OF ANY SUBSTANCE FROM THE CASING THROUGHOUT ITS LENGTH EXCEPT AT ENDS. CASING SHALL BE SO INSTALLED AS TO PREVENT THE FORMATION OF A WATERWAY UNDER THE RAILROAD, WITH AN EVEN BEARING THROUGHOUT ITS LENGTH, AND SHALL SLOPE TO ONE END.

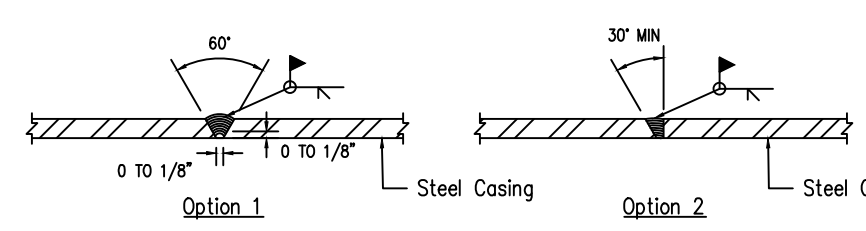
PVC CARRIER PIPE NOTES:

1. ALL RESTRAINED PVC PIPE SHALL BE AWWA C900-16 UL 1285 (MIN.).

Nominal Pipe Diameter	Minimum Nominal Casing I.D.
8"	16"
10"	18"
12"	20"
16"	24"
20"	36"



Steel Casing Pipe



Steel Casing Weld Detail (Typ)

NOTICE:
AQUA ADVISES THAT IT HAS COMPLIED WITH THE PROVISIONS OF THE UNDERGROUND UTILITY LINE PROTECTION LAW, ACT 287 OF 1974 AS AMENDED, IN PREPARING THESE DRAWINGS REQUIRING EXCAVATION OR DEMOLITION WORK AT SITES WITHIN THE COMMONWEALTH OF PENNSYLVANIA. AQUA DOES NOT MAKE ANY REPRESENTATION, WARRANTY, ASSURANCE OR GUARANTY THAT THE INFORMATION RECEIVED PURSUANT TO SAID ACT AND REFLECTED ON THESE DRAWINGS IS CORRECT OR ACCURATE OR THAT ALL SUBSURFACE UTILITIES AND STRUCTURES ARE SHOWN, BUT IS REFLECTING SAID INFORMATION ON THESE DRAWINGS IN ACCORDANCE WITH THE REQUIREMENT OF SUCH ACT. CONTRACTORS ARE REMINDED OF THEIR OBLIGATION TO NOTIFY 1-800-242-1776 OR 8-1-1 NOT LESS THAN 3 OR MORE THAN 10 WORKING DAYS PRIOR TO BEGINNING EXCAVATION.

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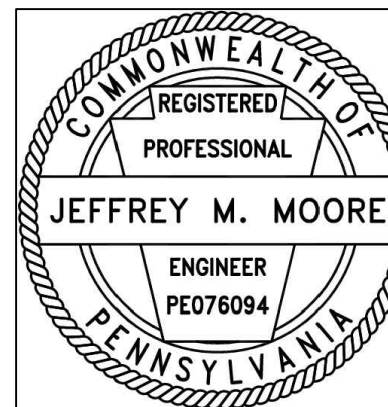
RIGHT-OF-ENTRY PERMIT PLAN

NO	DATE	REVISION	INTL
O		ISSUED FOR CONSTRUCTION	
O		DESIGN COMPLETION	

AQUA PENNSYLVANIA, INCORPORATED
762 LANCASTER AVENUE, BRYN MAWR, PA., 19010

PROJECT PLAN FOR:
LEE ST PROJECT
SEPTA RAILROAD CROSSING - JACK AND BORE
WHITEMARSH TWP, MONTGOMERY COUNTY

DRAWN BY: MD	CHK'D BY: JMM	EXT No: 19695-E
DATE: 5/16/2025	SCALE: N.T.S.	PLATE: K-17
PROJECT No: 466.21	CPA ID No: 8820387	A - 66967
APPROVED: <i>Jeffrey M. Moore</i>		SHEET 3 OF 11

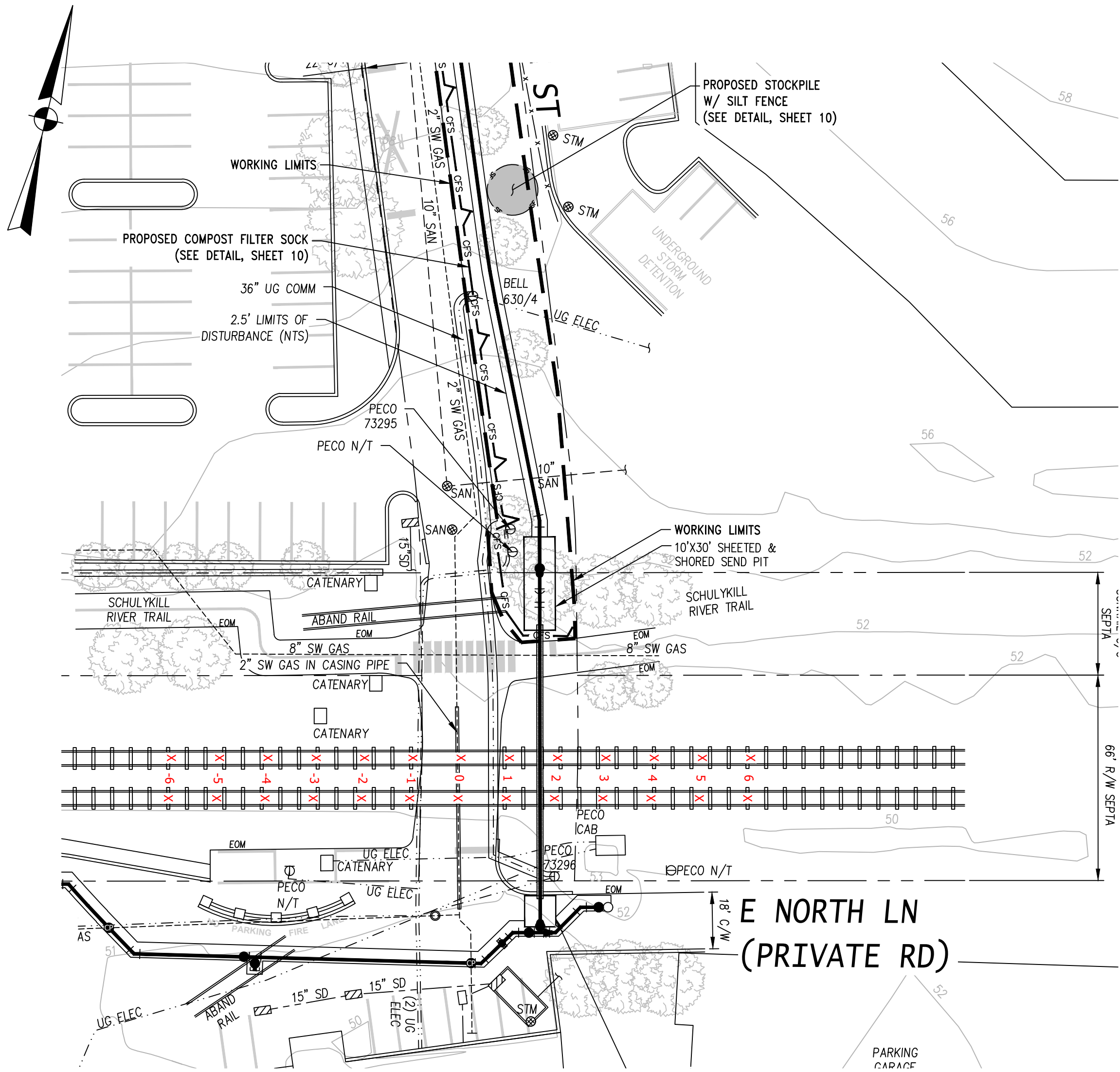


GENERAL NOTES:

- At a minimum, rail monitoring requirements are as follows:
1. Establish Remote Monitoring Points (RMPs) for collection of data.
 2. Preconstruction monitoring activities to begin at least 10 days prior to start of construction
 3. Monitor using total station instrument having a minimum angular accuracy of 1-second and an electronic distance measurement accuracy of 1.0mm + 2ppm. Total station to locate RMPs on track to be monitored
 4. Contractor shall space the prisms at 15'-6" intervals
 5. RMPs shall be either commercially available calibrated reflective targets or small prisms. All targets shall be mounted at a uniform elevation below the top of rail.
 6. Reflective targets shall be less than 3-inches square and mounted to the web of the rail by adhesive. Allowable minimum angle from instrument to target face is 30 degrees.
 7. Prisms mounted on rail brackets shall remain at least 1-inch below top of rail head and must not interfere with track components.
 8. Contractor shall include prisms on the track(s) the work passes beneath.
 9. If the prisms encounter special track work locations (crossings, turnouts, miter rails), Contractor shall notify SEPTA to determine an alternate arrangement.
 10. The total count along the track shall extend through the Zone of Influence and extend an additional five (5) stations, in either direction, outside of the Zone of Influence.
 11. Zone of Influence using the diameter or width of the underground work, extending to the surface of the ground at the soil angle of repose. Soil angle of repose should be taken from the depth of the underground work to ground level.
 12. Points shall be established 100 feet each direction of crossing at 15' 6" intervals on each rail per SEPTA ROW Design and Construction Standards, Section 9, subpart H.
 13. Points shall be set up at least one week prior to the start of excavation activities
 14. Initial readings must be taken at least two days prior to start of work
 15. During active construction readings shall be taken every hour
 16. During inactive construction, readings shall be taken every 12 hours
 17. After jack and bore completed and pits backfilled reading shall be taken once daily at same time of day for seven calendar days.
 18. Pre and post construction survey to be provided
 19. Readings to be hand delivered to SEPTA Engineer in charge onsite and emailed to SEPTA daily.
 20. Points shall be clearly marked in field on each rail both directions

RAIL MONITORING STATION KEY	
TARGET LOCATION	RAIL STATION
-6	
-5	
-4	
-3	
-2	
-1	
0	
1	
2	
3	
4	
5	
6	

RAIL MONITORING STATION KEY	
TARGET LOCATION	RAIL STATION
-6	
-5	
-4	
-3	
-2	
-1	
0	
1	
2	
3	
4	
5	
6	



RAIL MONITORING PLAN
SCALE - 1"=30'

2			JMM
O		ISSUED FOR CONSTRUCTION	
O		DESIGN COMPLETION	
NO	DATE	REVISION	INTL
AQUA PENNSYLVANIA, INCORPORATED 762 LANCASTER AVENUE, BRYN MAWR, PA., 19010			
PROJECT PLAN FOR: LEE ST PROJECT RAIL MONITORING PLAN WHITEMARSH TWP, MONTGOMERY COUNTY			
DRAWN BY:	MD	CHK'D BY:	JMM
DATE:	02/17/2022	SCALE:	N.T.S.
PROJECT No:	466.21	CPA ID No:	8820387
APPROVED		A - 66967	
		SHEET	11 OF 11

CM#

WO#

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